AY2026 Sem 1 BC2406/16 Computer Based Assessment (CBA)

Financial Loans Charged Off

Introduction

A Fintech provides loans to individuals and earns return from interest. The interest rate depends on the risk of the loan. At the end of the loan period, some applicants did not complete payments and will be charged off, taking a loss. The Fintech CEO hopes to use Machine Learning to reduce charged off.

Three datasets (trainset.csv, trainset.bal.csv and testset.csv) are provided. The meaning of the data variables are provided in Appendix A. The testset is meant to test the predictive accuracy of model and should not be used in model training or optimization. The outcome variable is loan_status.

Part A: Data Exploration and Data Preparation (30%)

1. What is the relationship between (a) grade and loan_status and (b) sub_grade and loan_status? Which of the two predictors – grade or sub_grade should be used in predictive models? Explain.
2. Explore the data using tables, statistics or charts and report your key findings.
3. Should term be treated as a continuous or categorical variable in predictive models? Explain.
4. Create a derived variable ITI (Installment to Income) defined as Annual Installment Payment to Annual Income ratio in the three datasets. Show the summary statistics (min, median, mean, max) of ITI in trainset.csv. Explain why ITI could be a potentially useful predictor of loan_status.

Part B: Predictive Models (30%)

Define Positive event if the model predicts loan_status = Charged Off. **Exclude id and grade from all models.** In logistic regression, set threshold as 0.5.

5. Using logistic regression on trainset.csv, show the testset confusion matrix. Comment on the results.

6. Show that trainset.bal.csv is balanced. Train the following models on trainset.bal.csv and display the testset False Positive Rate (FPR) and False Negative Rate (FNR) in a table (example table format provided below).

    i. Logistic Regression with all the remaining X variables. Complexity of logistic regression is the number of parameters to be estimated.

    ii. Optimised Logistic Regression with either p-value filter or/and Backward Elimination.

    iii. Maximal CART. Complexity of CART is the number of splits.

    iv. Optimised CART with either minimum 10-fold cross validation error or/and 1SE rule.

Model	Complexity	FPR	FNR
Log Reg (all X)	? parameters		
Log Reg (Optimised)	? parameters		
CART (max)	? splits		
CART (Optimised)	? splits		

Provide a brief commentary on the results to the Fintech CEO.

Part C: Recursive Feature Elimination (RFE) (20%)

Sinap (2024) and Chakravorty & Elsayed (2023)[1] showed that using RFE on Decision Tree (i.e. CART) and Random Forest can improve predictive accuracy significantly.

7. Apply RFE on CART and show the testset results. Did predictive accuracy improve significantly? Explain.

Part D: Advisory (20%)

8. Summarize your key findings and advise the Fintech CEO. [Max 200 words.]

[1] Both papers are provided in the assessment folder.

Appendix A

Variable	Description
id	Loan ID.
state	State of residence in USA.
emp_length	Employment length.
grade	Grade of the Loan as assessed internally by lender to evaluate credit risk. Grade A (Safest); Grades B, C, and D (Moderate Risk); Grades E, F, and G (High Risk).
home_ownership	Home ownership status.
loan_status	Loan status. Charge Off means the loan has written off unpaid debt as a bad-debt loss on their books after 120 days of missed payments.
purpose	Purpose for applying for the loan.
sub_grade	Each major letter grade from A to G is further divided into five numerical sub-grades (e.g., A1, A2, A3, A4, A5 down to G5). A1 represents the highest quality and least risky profile.
term	Term of the loan in months.
verification_status	Source Verified: Confirms authenticity by checking with the primary origin (e.g. employer, university) rather than relying on a third-party database; Verified: Confirms that the information matches a record against existing databases and secondary documents.
annual_income	Annual income.
dti	Debt-to-income ratio: The percentage of gross monthly income (before taxes) that goes toward paying monthly debt, such as rent, mortgages, student loans, and credit cards.
installment	Installment payment per month.
int_rate	Annnualised Interest Rate. Represents cost of borrowing to the borrower and return from lending to lender.
loan_amount	Loan amount.
total_acc	Total number of credit lines.